

Smart Plant Monitor and Automated Watering System Proposal

1. Abstract

This project presents the development of a Smart Plant Monitor and Automated Watering System designed specifically for indoor plants. The system uses an Arduino Uno R3 together with environmental sensors to monitor soil moisture, room temperature, and humidity in real time. A mini water pump controlled through a relay module automatically irrigates the plant whenever the soil moisture falls below a predefined threshold. The system aims to reduce human error in plant care and support users with busy schedules or frequent travel.

2. Objectives

- Accurately measure room temperature and air humidity around the plant.
- Monitor soil moisture levels using a capacitive soil moisture sensor.
- Automatically activate a mini water pump when soil dryness exceeds the threshold.
- Prevent overwatering by implementing intermittent pulse-watering logic.
- Monitor environmental conditions to identify potential risks of plant diseases and pest development.

3. System Architecture

The system operates through a simple monitoring and irrigation process:

1. The Soil Moisture Sensor and DHT11 sensor collect environmental data every 5 minutes.
2. The ESP32 processes incoming sensor data and evaluates environmental conditions.
3. The system evaluates temperature, air humidity, and soil moisture conditions to identify environments that may encourage fungal diseases, mold growth, or pest infestations.
4. If the soil moisture level falls below approximately 35%–40%, the relay module is activated automatically until the soil moisture reaches about 60%.
5. The relay supplies power to the 5V mini water pump, which delivers water directly to the plant pot.

6. Once the soil moisture reaches the predefined optimal level, the pump is automatically turned off to prevent overwatering.

4. Hardware Implementation

4.1 Components List

Component	Quantity	Function
ESP32	1	Main microcontroller for processing sensor data
Capacitive Soil Moisture Sensor v1.2	1	Measures soil moisture
DHT11 Temperature and Humidity Sensor	1	Measures room temperature and humidity
5V 1-Channel Relay Module	1	Controls water pump activation
5V Mini Water Pump	1	Supplies water to the plant
1m Pipe (6 mm)	1	Transfers water to the pot
Breadboard MB-102	1	Prototype connections
Male-to-Female Jumper Wires	1	Electrical connections
Male -to-Male Jumper Wires	1	Electrical connections

5. Methodology

- **Data Collection:** reads analog data from the soil moisture sensor and digital data from the DHT11 sensor every 5 minutes.
- **Decision Making:**
 - The system checks whether soil moisture falls below the dry threshold.
 - The system also evaluates temperature and humidity levels to estimate potential environmental risks for plant health.
 - Normal Condition (Low Risk)

When the temperature is between 18°C and 30°C, the air humidity is between 40% and 70%, and the soil moisture is between 30% and 60%, the plant is considered to be growing under healthy conditions with a low risk of disease or pest infestation.
 - Mold Growth Risk (Medium Risk)

When the temperature is between 20°C and 30°C, the air humidity exceeds 80%, and the soil moisture is above 70%, the environment becomes favorable for mold development. Excess moisture prevents the soil from drying properly, increasing the likelihood of fungal growth on the soil surface and leaves.

- **Root Rot and Fungal Disease Risk (High Risk)**

When the temperature is between 22°C and 30°C, the air humidity exceeds 85%, and the soil moisture remains above 80% for an extended period, the plant faces a high risk of root rot and fungal infections. Common symptoms include yellowing leaves, soft stems, and darkened roots.

- **Pest Infestation Risk (Moderate to High Risk)**

When the temperature ranges from 25°C to 32°C, the air humidity is between 70% and 90%, and the soil moisture exceeds 75%, the conditions become suitable for pests such as mealybugs and fungus gnats, which may damage the plant and contribute to the spread of plant diseases.

- Warning messages can be generated when conditions become favorable for fungal diseases, mold growth, or pest development.

- **Execution:** The relay activates the mini water pump to water the plant until the optimal moisture level is reached.

6. Software Implementation

- Arduino IDE is used for programming the ESP32 using C++ language.
- The ESP32 continuously reads data from the Soil Moisture Sensor and DHT11 sensor.
- Sensor values are processed every 5 minutes.
- The ESP32 connects to a Wi-Fi network and uploads sensor data to an IoT cloud platform such as ThingSpeak or Blynk.
- Users can remotely monitor temperature, humidity, soil moisture, and environmental risk conditions through an Internet-connected dashboard.

7. AI Model Development

This project does not utilize artificial intelligence or machine learning models. Instead, the system relies on sensor-based automation and predefined threshold logic implemented directly on the ESP32 microcontroller.

The moisture threshold for the plant is calibrated experimentally to ensure the plant receives sufficient water while preventing excessive watering.

8. Model Performance & Results

- The system successfully monitors room temperature, humidity, and soil moisture in real time.
- The mini water pump automatically activates whenever the soil becomes too dry.
- Pulse-watering logic helps reduce the risk of root rot and soil waterlogging.
- The prototype is expected to maintain healthy plant conditions during long weekends or short vacations.
- The system can provide environmental risk warnings when temperature and humidity conditions become favorable for plant diseases or pest development.
- Environmental data can be monitored remotely through an IoT dashboard.
- Historical sensor data can be stored and visualized using cloud-based monitoring platforms.

9. System Testing

- Verify all hardware components are connected correctly.
- Power on the Arduino Uno and sensors.
- Observe sensor readings through the Serial Monitor.
- Dry the soil slightly and confirm that the relay activates the water pump.
- Check whether the pump stops automatically after the optimal moisture level is reached.
- Run the system continuously to evaluate stability and reliability.
- Verify Wi-Fi connectivity and successful cloud data transmission.
- Confirm that sensor readings are updated correctly on the IoT dashboard.

10. Troubleshooting

Soil Moisture Sensor Not Responding

Check wiring connections and verify the analog pin configuration.

Pump Not Working

Ensure the relay module receives sufficient power and confirm correct relay wiring.

Incorrect Sensor Readings

Recalibrate the sensor threshold values and inspect sensor placement.

Arduino Not Uploading Code

Verify USB cable connection, COM port selection, and Arduino IDE configuration.

Internet Connection

Verify Internet connectivity, API keys, and dashboard configuration

Wifi connection failure

11. Conclusion

The Smart Plant Monitor and Automated Watering System provides an effective and affordable solution for maintaining indoor plants automatically. By integrating environmental sensors, Arduino-based control logic, and an automated watering mechanism, the system reduces manual effort and ensures healthier plant growth. In addition to automated irrigation, the system can be extended to monitor environmental conditions and provide early warnings about potential disease or pest risks. This demonstrates how low-cost IoT-based smart plant monitoring and automated irrigation system can improve plant care, convenience, and preventive plant health management. By integrating environmental sensors, ESP32-based control logic, automated irrigation, and cloud connectivity, the system enables real-time monitoring and remote access to plant health information. The project demonstrates how low-cost IoT technology can improve plant care, convenience, and preventive plant health management.

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